

196. Delete Duplicate Emails

PLATFORM

Platform: LeetCode

DIFFICULTY

EASY

DATE

Solved: June 3, 2026

Problem

Delete duplicate email records from the `Person` table.

- Keep only **one copy** of each email.
- Keep the row with the **smallest id**.
- Delete all other duplicate rows.

Execution Flow

- Join `Person` with itself.
- Match rows having the same email.
- Find the row with the larger `id`.
- Delete that row.
- Keep the row with the smallest `id`.

Concepts Used

1. Self Join

A table is joined with itself.

```
FROM Person p1  
JOIN Person p2
```

- `p1` and `p2` are aliases for the same table.
- This allows us to compare rows within the table.

2. Find Duplicate Emails

```
p1.email = p2.email
```

This condition matches rows having the same email.

3. Identify Which Row to Delete

```
p1.id > p2.id
```

For duplicate emails:

- Smaller `id` is kept.
- Larger `id` is deleted.

The screenshot displays the LeetCode problem page for "196. Delete Duplicate Emails". The problem is marked as "Solved" and "Easy". The description states: "Delete duplicate email records from the `Person` table. Keep only one copy of each email. Keep the row with the smallest `id`. Delete all other duplicate rows." The `Person` table schema is shown with columns `id` (int) and `email` (varchar), where `id` is the primary key. The solution is provided in MySQL:

```
1 # Write your MySQL query statement below  
2 DELETE p1  
3 FROM Person p1  
4 JOIN Person p2 ON p1.email = p2.email  
5 WHERE p1.id > p2.id
```

 The runtime performance is shown as 911 ms, beating 27.58% of submissions. The interface also includes a "Testcase" section showing the input and output for the first case.

196. Delete Duplicate Emails Solved

Easy Topics Companies

SQL Schema Pandas Schema

Table: `Person`

Column Name	Type
<code>id</code>	<code>int</code>
<code>email</code>	<code>varchar</code>

`id` is the primary key (column with unique values) for this table. Each row of this table contains an email. The emails will not contain uppercase letters.

Write a solution to **delete** all duplicate emails, keeping only one unique email with the smallest `id`.

For SQL users, please note that you are supposed to write a **DELETE** statement and not a **SELECT** one.

For Pandas users, please note that you are supposed to modify `Person` in place.

After running your script, the answer shown is the `Person` table. The driver will first compile and run your piece of code and then show the `Person` table. The final order of the `Person` table **does not matter**.

The result format is in the following example.

Example 1:

Input:
`Person` table:

<code>id</code>	<code>email</code>
1	john.doe@company.com
2	john.doe@company.com
3	john.doe@company.com
4	john.doe@company.com
5	john.doe@company.com

Accepted 22/22 testcases passed
Sahil Khan submitted at Jun 03, 2026 13:55

Runtime
911 ms | Beats 27.58%

Code | MySQL

```
1 # Write your MySQL query statement below  
2 DELETE p1  
3 FROM Person p1  
4 JOIN Person p2 ON p1.email = p2.email  
5 WHERE p1.id > p2.id
```

More challenges

- 1731. The Number of Employees Which Report to Each Employee
- 3278. Find Candidates for Data Scientist Position II
- 3465. Find Products with Valid Serial Numbers

Write your notes here

Testcase | Test Result

Accepted Runtime: 74 ms

Case 1

Input

`Person` =

<code>id</code>	<code>email</code>
1	john.doe@company.com
2	john.doe@company.com
3	john.doe@company.com
4	john.doe@company.com
5	john.doe@company.com